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CIA-1

1. Given

$p(c) = 0.03$, $p(+ | c) = 0.90$, $p(- | c) = 0.94$

λ (no | c) = 50, λ (treat | - c) = 2

1. priors and likelihoods:

$p(-c) = 0.97$, $p(+ | c) = 0.90$, $p(+ | -c) = 0.06$, $p(- | c) = 0.10$

$p(- | -c) = 0.94$

The false-positive rate is the complement of specificity.

2. Evidence: $p(+) = 0.90 \times 0.03 + 0.06 \times 0.07$

$= 0.0582$

$p(-) = 1 - 0.0852 = 0.9148$

Total probability combines the deseased and healthy population.

3. posteriors: $p(c | +) = 0.0270 / 0.0852 = 0.3169$

$p(-c | +) = 0.6831$, $p(c | -) = 0.10 \times 0.03 / 0.9148 = 0.0033$

Even after a positive test the probability of disease is only about 31.7%

4. conditional risk:

R (treat | +) $= 2 \times 0.6831 = 1.3662$, R (no treat | +) $= 50 \times 0.3169$

$= 15.845$

R (treat | -) $= 2 \times 0.9967 = 1.9934$ R (no treat | -) $= 0.1640$

conditional risk $R(\alpha | x) = \sum \lambda(\alpha | w)(p(w | x)$

5. Minimum-risk decision:

positive → treat (1.3662 < 15.845)

negative → do not treat) (0.1640 < 1.9934)

$p(c | x) > 2/(2+50) = 0.0385$

$p(c | +) = 0.3169$ exceed it

$p(c/L) = 0.0033$ does not

6. Bayes risk:

$R = 0.0852 \times 1.3662 + 0.9148 \times 0.1640 = 0.2664$

$2 \times p(-c, +) \pm 50 \times p(c, -) = 2 \times 0.0582 + 50 \times 0.0030$

$= 0.2664.$

7. second independent positive test:

posterior odds $= (0.3169 / 0.6831) \times (0.90/0.06)$

$= 0.4639 \times 15 = 6.959$

$p(c/++) = 6.959/7.959 = 0.8744$

The previous posterior becomes new prior

8. Minimum error (0-1 loss) rule:

$p(c/x) > 0.5$ since $p(c/+) = 0.3169 < 0.5$

The rule never exist, its expect cost under

The real loss matrix $= 50 \times 0.03 = 1.5$

The minimum risk rule reduces expected cost from 1.5 to 0.2664, a reduction of 82.84%.

conclusion: A positive result gives only a 31.7% change of disease, yet treatment is justified because a missed case costs 25 times a false alarm.

2. Given

load x	Y = NO	Y = Yes	P(x)
low	0.30	0.02	0.32
Medium	0.28	0.10	0.38
High	0.10	0.20	0.30
P(y)	0.68	0.32	1.00

1. Marginals: $p(x) = (0.32, 0.38, 0.30)$

$p(y) = (0.68, 0.32)$

Row and column sums of joint table

2. $H(x) = -\Sigma p \log_2 p = 0.5260 + 0.5304 = 1.5776$

$H(y) = 0.3784 + 0.5260 = 0.9044$ bits

Each term is $-p \cdot \log p$ eg $-0.32 \log_2 0.32 = 0.5260$.

3. $H(x, y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.3322$

$+ 0.4644 = 2.2770$ bits

Sum of over all six joint probability

4. $H(y/x) = H(x, y) - H(x) = 2.2770 - 1.5776 = 0.6994$ bits

chain rule of entropy.

5. $I(x:y) = H(y) - H(y/x) = 0.9044 - 0.6994 = 0.2050$

$I(x:y)$ is 22.7% of $H(y)$

6. cross entropy $H(p \times Q) = -\Sigma p \times \log_2 Q = 0.32$

$(1.000) + 0.38 (1.7370) + 0.30 (2.3219) = 1.6766$ bites

Average code length when data from p x are coded

7. KL divergence:

$D_{KL}(p \times \| Q) = H(p \times Q) - H(x) = 1.6766 - 1.5776$

$= 0.0990$ bits.

10

Always ≥ 0.

8. $p(y = yes \mid x = High) = 0.20 / 0.30 = 0.6667$

fault probability rises sharply with load

compared with overall $p(yes) = 0.32$

Conclusion: The mutual information of 0.205 bits

is positive, so load carries useful but only moderate information about fault (high load 66.7% fault probability versus 6.25% at low load).

3. i) Given

samples $x = \{48, 52, 55, 47, 53, 51\}$, $n = 6$

1. ML mean: $\mu = (48+52+55+47+53+51)/6$

$= 51.0°C$

The ML estimate of Gaussian mean is sample mean.

2. squared deviation:

$(-3)^2 + 1^2 + 4^2 + (-4)^2 + 2^2 + 0^2 = 9+1+16+16+4+0 = 46$

ML variance $\sigma^2 = 46/6 = 7.667 \Rightarrow \sigma = 2.769°C$

3. unbiased variance

$s^2 = 46/5 = 9.20 \Rightarrow s = 3.033°C$

Dividing by n-1 corrects the bias

because $E(\sigma^2, ML) = ((n-1)/n)\sigma^2 = 5/6(\sigma^2)$

4. $p(x > 55)$, $z = (55-51)/2.769 = 1.445$

$p = 1 - \Phi(1.445) = 0.0743$

5. $p(48 < x < 54)$: $z = \pm(3/2.769) \pm 1.083$

$p = 2\Phi(1.083) - 1 = 0.7214 \approx 0.72$

Accept 0.72 ± 0.01 using $\Phi(1.08) = 0.8599$

conclusion:- ML estimates are $\mu = 51.0°C$ and $\sigma^2 = 7.667°C^2$. The unbiased variance is larger 9.20 for small samples ML variance the true

variance and therefore underestimate the probability (0.0743 against 0.0936 for $P(X>55)$).

ii) Given

$(x, y) = (10,25), (20,38), (30,52), (40,61), (50,79)$

$\lambda = 500$

1. Means: $\bar{x} = 30$, $\bar{y} = 25+38+52+61+79 / 5 = 51$

$S_{xx} = \Sigma (x-\bar{x})^2 = 400+100+0+100+400 = 1000$

$S_{xy} = \Sigma (x-\bar{x})(y-\bar{y}) = 520+130+0+100+560 = 1310$

2. $b_1 = 1310/1000 = 1.31$, $b_0 = 51 - 1.31 \times 30 = 11.7$ ms

fitted model $\hat{y} = 11.7 + 1.31x$.

3. fitted values: 24.8, 37.9, 51.0, 64.1, 79.2

$SSE = 0.04 + 0.01 + 1.00 + 9.61 + 3.24 = 13.90$

SSE measures the unexplained variation

4. $SST = 676 + 169 + 1 + 100 + 784 = 1730$

$R^2 = 1 - (13.90/1730) = 0.9920$

5. prediction at $x = 60$ $\hat{y} = 11.7 + 1.31 \times 60 = 90.3$ ms

$x = 60$ lies outside the training range 10–50,

so this is extrapolation

6. Ridge: $b_1 = S_{xx}/(S_{xx}+\lambda) = 1310/1500 = 0.8733$

$b_0 = 51 - 0.8733 \times 30 = 24.8$

$\hat{y}(60) = 24.8 + 52.4 = 77.2$ ms

$SSE = 204$, $R^2 = 0.8817$

(2) shrinks the slope by 33.3%

conclusion: almost perfectly linear points

$\lambda = 500$ over regularises and underfits (R^2 drops from 0.992 to 0.882) biasing the 60% prediction down to 77.2ms.

4. Given

N-spam = 40, N-legit = 60

Tp = 45, fN = 15, fp = 10, TN = 130

1. priors: p(spam) = 40/100 = 0.40

p(legit) = 60/100 = 0.60

class frequies in train set

2. smoothed likelihoods, spam (denominator 42)

offer = 29/42 = 0.6905 free 25/42 = 0.5952

meeting 5/42 = 0.1190

Adding 1 to the numerator and 2 to the denominator avoids zero probability.

3. smoothed likelihoods.

offer 7/62 = 0.1129, free 4/62 = 0.0645.

meeting 34/62 = 0.5484

The word "meeting" is typical of legitimate mail

4. scores: x = (offer=1, free=1, meeting = 0)

spam = 0.40 x 0.6905 x 0.5952 x (1-0.1190) = 0.1448

legit = 0.60 x 0.1129 x 0.0645 x (1-0.5484) = 0.00197

5. posterior

$p(spam|x) = 0.1448 / (0.1448 + 0.00197) = 0.9866$

$p(legit|x) = 0.0134$

6. cost-sensitive threshold

$p(spam|x) > p(legit|x)$

$p(spam|x) > 5/6 = 0.833$

$0.9866 > 0.833 \rightarrow$ classify as spam

7. Accuracy: $(45+130)/200 = 87.5\%$

precision $= 45/(45+10) = 81.82\%$

Recall $= 45/(45+15) = 75\%$

$f_1 = 2TP / (2TP + fp + fn) = 90/115 = 0.7826$

8. Total misclassification cost

$5 \times fp + 1 \times fn = 5 \times 10 + 15 = 65$

false-positive rate $= 10/140 = 7.14\%$

Conclusion:- The new email is classified as spam with posterior 0.9866 well above the 0.833 cost sensitive threshold

$x = (0.5, 1.0)$ $t = 1$ $\eta = 0.5$

$w_{11} = 0.4$, $w_{21} = -0.2$ $b_1 = 0.1$, $v_{12} = 0.3$

$w_{22} = 0.5$, $b_2 = -0.1$ $v_1 = 0.6$

$v_2 = 0.4$, $b_3 = 0.2$